

Rearrange Social Overloaded Posts to Prevent Social Overload

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ABSTRACT

According to the latest investigation, there are 1.7 million active social network users in Taiwan. Previous researches indicated social network posts have a great impact on users, and mostly, the negative impact is from the rising demands of social support, which further lead to heavier social overload. In this study, we propose social overloaded posts detection model (SODM) by deploying the latest text mining and deep learning techniques to detect the social overloaded posts and, then with the developed social overload prevention system (SOS), the social overload posts and non-social overload ones are rearranged with different sorting methods to prevent readers from excessive demands of social support or social overload. The empirical results show that our SOS helps readers to alleviate social overload when reading via social media.

KEYWORDS

deep learning, sentiment analysis, social overload, social support seeking behavior, rearrangement

1. INTRODUCTION

In the age of big data, social network users frequently receive abundant messages and to some degree are compulsory to engage in them via different social media channels. This may not only cause information overload, but previous research [1] also has revealed that the more often the users use social media such as Facebook, the more depressed they are. The reason is that these users are often exposed to negative posts such as passive-aggressive posts, pity-seeking posts, attention-craving posts, narcissistic posts, and so on.

When the messages are mixed up with seeking more needs of social support, people may encounter social overload. Numerous psychology studies [2, 3] indicated that those who write to post their updates or seek social supports online could temporally lighten their psychological load while at mean time transfer their load to their readers and cause them to feel stressful and depressed. The aforementioned phenomenon is known as social support seeking behavior.

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According to the latest investigation, social network leads as the top online activity and more people tend to seek support from the Internet instead of from the professionals when they encounter difficulties unable to be solved by themselves in their lives. Thus, a growing number of people are suffering from Internet depression. When more and more people seek social support from social networks, each user also receives increasing requests of social support. In other words, awareness of other people's problems is associated with a range of negative outcomes, such as depression- that is what Keith Hampton said, the "cost of caring". Robin Dunbar indicated that if our Facebook or Twitter friends repeatedly request social support, we will soon feel emotionally exhausted [4], or social overload. Social support originally serves as a positive

interaction mechanism among users; however, without considering social overload, the existing first-in-first-out page display or view count label of display on social network sites may result in the delivery of massive negative messages at a time. Readers who repeatedly receive these would experience more stress and further suffer from more psychological disorders.

In this study, we propose an innovative social overload prevention system (SOS) to adequately label the social overload posts in the existing social network sites by the psychological points of view to reduce the negative impact on the readers. To date, there is still little research discussing auto-prediction and treatment of social support seeking behavior, much less the discussion of social overload prevention. Thus, we develop social overloaded posts detection model (SODM) to cope with this problem. This SODM includes data collection, deep learning algorithms, machine learning, and text mining algorithms which assist computers to understand human languages and extract social overload messages. The empirical results show SOS helps to reduce the labor cost for analyzing sentiments in posts and achieves 95.15% accuracy in predicting social overload messages. By rearranging the order of social overload posts, 75% participants report the stress caused by social overload is reduced.

2. RELEATED WORKS

Various deep learning models have been used in sentiment analysis [5-8]; for example, pre-trained word vectors are achieved by applying neural networks [9] and the word vectors pertained from [9] are loaded to word embedding layer of neural networks [5]. Then with the use of convolutional neural network (CNN) or recurrent neural network (RNN) the model developed by [5] performs superior classification to the use of SVM-based approaches and Paragraph to vector (P2V) [10]. In this study, we only discuss the methods proposed by selected studies [5-7] and their improvements rather than the previously used SVM and P2V.

Numerous studies have proposed different model architectures for pre-trained word vector. Many of them compare the use of continuous bag-of-words (CBOW) [9] with it of word-word co-occurrence [11] and many results show that both of them are able to achieve good word pre-training and each has its strength. The Skip-Gram model, predicting the context given a word, is considered the inverse of CBOW and has demonstrated better training result [5]. In this study, we adopt the method proposed by [9] for the reason of better training speed and we follow [5] to apply Skip-Gram instead of CBOW.

3. METHODOLOGY

We develop SODM with machine learning and text mining methods in this study, and design algorithms of rearrangements to prevent users to be social overloaded for our SOS. Its model architecture, is composed of five layers: sentiment resource, compare neural network model, the most accurate matching model and rearrange methods.

We, first, apply the previous technique [9] to train collected data sets to obtain word vectors. Then, the word vectors are loaded to the embedding layer of neural network. Then, we wonder whether there is overlapping or repeated sentiments of hatred and melancholic, so after the words and sentences are sorting, they are classified by two methods proposed in the present study. The first one is to assign data from Prozac and Hate board to the social overload category while using data from other boards as the classified normal overload category. The other one is to classify data into three categories according to the three sources: Prozac, Hate or others. Since the order of articles is posted based on time series, the amount of posts collected from different boards might be different. To avoid the imbalance data of different boards caused by our classification, we extract the same number of posts from each category for learning. For the first proposed method, 200 thousand of posts are extracted from each of Prozac and Hate, while 400 thousand of posts are from others. As for the latter method, 300 hundred of posts are extracted from Prozac, Hate, and others each. Then all the extracted articles are mixed up to prevent the mathematical model to learn a specific sentiment merely in the same batch, which usually occurs when tremendous posts from a specific board are presented at a time.

We adopt some latest sentiment analysis techniques based on different types of deep neural networks to develop our mathematical model [5-7, 10]. The baseline in this study adopts the approach proposed in [10] using a CNN based model with 2D filters and Max-Pooling to extract features in both sentence level and document level. K-Max-Pooling [6], a more efficient pooling, is introduced and applied in both sentence level and document level in [7]. It extracts TOP-K units to serve as the parameters for sentence level classification; meanwhile to preserve more missing information of max-pooling, as shown in. [5] performs state-of-the art classification by using Gated Recurrent Neural Network (GRNN) and Long-Short Term Memory (LSTM). [5] proposed two approaches. One is to adopt CNN in sentence level and GRNN in document level. The other one is applying LSTM in sentence level and GRNN in document level. In

previous studies, pooling is performed at the sentence level. Due to the grammar errors or typos, we speculate the coherence between the end and the beginning of the two adjacent sentences might be weak. Therefore, in our proposed CNN-K-max-Pooling in document level with GRNN (CKDGNN), we first concatenate all sentence embedding as document embedding and processed by K-Max-Pooling, then, include time-series flatten to store the time series output data from K-Max-Pooling layer. Furthermore, we adopt the approach proposed in [5] to connect GRNN and concatenate with various-sized filters for pooling [7](Fig .5). Since our pooling model is more global, the concept of filter utilization is adopted from [8] while the filter's width is multiplied. Three models, static, non-static and multi-channel models, are presented in [7] and differentiated by their ability to fine-tune the pre-trained word vectors [7]; then, they are applied to datasets with different features and the conclusion of [7] indicates the performance of three models has no remarkable difference. Inspired by [5-9, 12, 13], we present our new model CKDGNN for social overload detection, and we compare our model with DCNN [6], CNN-GRNN [5], LSTM-GRNN [5]. As for the last output layer, our model uses the commonly practiced Softmax activation function for the output of sentiments because it normalizes the sum of sentiment values.

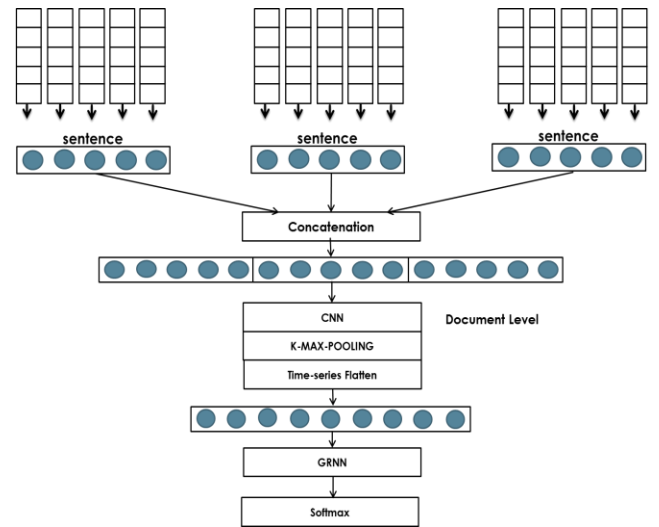


Fig. 1. CNN-K-max-Pooling in document level with GRNN (CKDGNN).

After the neural network is built, gradient descent is used in the training to achieve the optimization and to approach the local minimum of the function. However, there are various algorithms to optimize gradient descent and each of them greatly affects the rate of convergence as well as the required accuracy of a model [14]. Nadam (Nesterov-accelerated Adam) has stronger bound in learning, more direct effect on gradient update compared with other methods [15].

Early stopping [16], widely used in training neural networks. We also adopt unsupervised pre-training since the experiments in [17] confirm and clarify the use of unsupervised pre-training helps to improve robustness of deep neural networks.

We randomly extract 300 posts from PTT to predict the probability of social overload. Those reposted and shared posts are deleted to exclude the mistaken social support requesting from journalists or the press. Then, we use our proposed model to compute the sentiment score of each classification directly. We proposed two methods for classification: one is classified with three sentiments, and the other one is with two sentiments. Next, we sort the normal load posts according to the probability calculated by our DNN. They are sorted from the highest probability to the lowest probability. Then, all the rearranged posts are divided into two groups and further cross arranged to generate a new order of posts. With this, the sentiment score of a successive dataset is more stable and posts with higher probability of social overload are prevented from presenting together for over half of the dataset; however, we still decide to employ threshold sort because the aforementioned cross arrangement dramatically alter the original order of the posts in the dataset and presumably, people's tolerance of social overload is higher than expected. We define 0.5 as our threshold to identify the social overloaded posts and three posts with high probabilities of social overload as the maximum tolerance in SOS. If three posts of high probabilities of social overload are successively adjacent, SOS will insert one post with threshold lower than 0.5 behind these three posts. To sum up of our SOS, we use two methods of classification, five deep neural networks for sentiment analysis and four approaches for detecting probabilities of social overload in posts, and 50 students participated in this empirical study.

4. RESULTS AND DISCUSSION

There are five types of social support [4, 18], and among them, emotional support is the most commonly perceived when we hear of it. Emotional support offers to mitigate an individual's negative sentiment, most often related to melancholia and complaints. Previous studies [19, 20] extracted only messages related to melancholia from NTU's Prozac board. In this study, we extract messages related to melancholia and messages related to complaints from NTU's Prozac and Hate board, respectively. Each PTT board has its moderator and regulations. The moderator or administrator of a PTT board is entitled to delete inappropriate posts. Thus, most of spams or nonsense posts decrease. In addition to extracting data from NTU's Prozac and Hate boards, we also extract data from other boards to allow our model to learn other types of sentiments apart from melancholia and complaints as well as to avoid the overfitting resulted from imbalanced data which may further cause difficulties in determining nonsocial support behaviors. This design allows more balanced data for our model and achieves better learning results.

There are over 1.5 million registered users and 20 thousand boards with various themes on NTU's BBS. We use a web crawler to extract posts from the top 5 popular boards before October of 2016 and posts from top 1500 pages of Prozac and Hate board; in total, 5.07 million posts are extracted initially in our experiment.

We adopt 5-fold cross-validation in our experiment. The

accuracy of the two types of classification is presented in Tables I and II. The result indicates that the connected model of Word2vec vector combined with CKDGNN outperforms others in sentiment classification. The possible reason for this result is that the extracted features from CNN K-Max-Pooling in sentence level weaken the coherence of the end and the beginning of the two adjacent sentences. It also leads to the result that sentence coherence cannot be learned when GRNN is used in document level.

TABLE I. THE ACCURACY OF THE FIRST METHOD OF CLASSIFICATION (5-FOLD CROSS-VALIDATION).

Method	Accuracy
DCNN [12]	0.9072
CNN-GRNN [11]	0.9088
LSTM-GRNN [11]	0.9040
CKDGNN	0.9515

TABLE II. THE ACCURACY OF THE SECOND METHOD OF CLASSIFICATION (5-FOLD CROSS-VALIDATION).

Method	Accuracy
DCNN [12]	0.8930
CNN-GRNN [11]	0.8763
LSTM-GRNN [11]	0.9038
CKDGNN	0.9290

Prior to the cross arrangement of the data, auto encoding is employed in neural network pre-training [21] to improve the reliability and the result shows that the classification accuracy of the proposed CKDGNN model is elevated to a 0.9653 of accuracy in sentiment analysis. This trained model is further deployed to obtain each post's probabilities of social overload and based on the given probability for our SODM, the posts are rearranged and a new post order is generated as well. The different post sorting methods are in our experiments.

Then, these different post sorting methods are compared with the original order of posts by the participants for the purpose of realizing whether these different arrangements of posts would help to relieve their social load or would let them stop at a specific one with unbearable social overload by our SOS.

In the last stage of our experiment, we implement our SOS and make participants read posts with different post sorting methods and based on their reported results, they are divided into two groups: *preference* and *finishing reading*, as shown in Table III. If participants feel less stressed during the reading with one specific post sorting method, they are classified to the *preference* group. The value of *preference* is obtained by "the numbers of people who feel less stressed" being divided by "all participants". *Finishing reading* is defined as those who don't finish reading in a specific post sorting method and its value is obtained by "the number of participants finishing reading with a specific arrangement" being divided by "all participants". Most participants report they feel less stressed with cross-rearrangement and threshold-rearrangement has been proved to surpass other post sorting methods with respect to social overload. The proposed threshold-rearrangement, although is not the most effective to prevent social overload, yield satisfied results. The given reasons for this result might be associated with the individual differences of social

overload tolerance and the high threshold value set in the present work. However, cross-rearrangement is not very efficiency and cost-effective in sorting because almost all posts need to be rearranged

TABLE III. THE EMPIRICAL EXPERIMENT RESULT

Article Sorting	Preference (%)	Finishing Reading (%)
Original	20	50
Cross-Rearrangement	40	75
Threshold Rearrangement	30	60

Compared with cross-rearrangement, threshold-rearrangement yields a more efficient performance; thus, it successfully rearranges the order of partially successive social overloaded posts and lets readers feel relieved. There are **41** appearances of three successive social overload posts in the original post sorting order, as shown in; however, after the threshold-rearrangement, no successive social overload posts are found. Also, threshold-rearrangement is easy to be implemented in social networks.

5. CONCLUSION

Previous research has proved that various social support behaviors cause social overload. In this study, we develop SODM to detect and rearrange social overload posts from Prozac and Hate boards. After comparing with several latest deep neural networks utilizing in sentiment analysis, we proposed CKDGNN which yields the best result in extracting the melancholic and hatred sentiments from the datasets. In this work, we hypothesize if posts are rearranged, the successive display of social overload posts will be avoided. After experimenting different post sorting methods, our hypothesis is evidenced and the results show that by breaking the successive display of social overload posts, participants' stress decrease. Furthermore, if the most suitable threshold value and numbers of tolerance posts can be found for each individual, we will be able to optimize the posts arrangement with the least alteration of the original post display order.

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